

Elbert Chao

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EDUCATION

Western University

London, ON

Master of Engineering Science (MEngSc), Software Engineering

Sept. 2025 – Expected Apr. 2027

Thesis: Uncertainty Quantification for Deep Learning in Medical Imaging — **CGPA: 4.0/4.0**

Awards: Ontario Graduate Scholarship (\$15,000); Med-i CREATE Scholarship (\$10,000)

Coursework: Digital Image Processing, Dependable AI Systems, LLM/VLM, Data Analytics

TECHNICAL SKILLS

Programming Languages: Python, TypeScript

ML & AI: PyTorch, TensorFlow Lite, OpenAI API, LangChain, Transformers, Recommender Systems, Computer Vision

Backend & Data: FastAPI, Flask, MongoDB, REST APIs

Tools: GCP, Streamlit, Nginx, Jira, Confluence

EXPERIENCE

Pinterest

May 2026 – Aug. 2026

Machine Learning Intern (PyTorch, Transformers, RecSys, Distributed ML)

Toronto, ON

- Enhanced **Homefeed candidate retrieval quality** by developing sequential **Transformer-based user-interest embeddings** from interaction logs and implementing production diagnostics for latency, overlap, and interest clustering.
- Separated engaged Pins from similar non-engaged content by formulating and tuning a **margin-based pairwise ranking loss**, identifying it as the strongest driver of **offline engagement metric lift** across candidate sets.
- Validated offline retrieval gains into live user engagement by deploying model artifacts and orchestrating **controlled online A/B experiments** while systematically auditing **serving complexity vs. latency tradeoffs**.

PM Accelerator

Jun. 2025 – Aug. 2025

AI Engineer Intern (Python, FastAPI, OpenAI API, MongoDB)

Remote (Boston, MA)

- Built a semantic resume to job matching system using **OpenAI API** and **FastAPI**, improving alignment accuracy by **35%**.
- Developed a Python web scraping and text normalization pipeline to ingest unstructured job data, reducing manual processing by **60%**.
- Designed a scalable **RESTful API** to support resume parsing and context-aware interview question generation.

Free Appropriate Sustainability Technology – FAST

Nov. 2024 – Apr. 2025

Full-Stack Developer (Next, Python, Flask, Nginx, PM2)

London, ON

- Developed a full-stack solution with a **Next frontend (TypeScript)** and **Flask backend (Python)** to deliver ML-based insights into strawberry health, resulting in a **90%** increase in crop monitoring effectiveness.
- Enhanced AI model performance by conducting real-world tests, systematically debugging edge cases and refining feature inputs, achieving classification speeds under **500ms**.
- Maintained server uptime across a **2-month** period with **fewer than 3** total downtime incidents using Nginx and PM2 for load balancing and process management.
- Configured and secured a production environment for a campus-deployed web service, enabling **24/7** availability throughout the term with minimal downtime or configuration drift.

PROJECTS

Probabilistic Deep Learning for Kidney Ablation Support

Aug. 2025 – Present

Graduate Researcher (Python, PyTorch, CNNs, Transformers)

- Refined a **hybrid CNN–Transformer** segmentation model with custom stratified resampling to address severe data scarcity, improving Dice score by **27%** (58% → 85%).
- Implemented **Monte Carlo Dropout** and **layer-wise uncertainty heatmaps** to quantify epistemic uncertainty, producing pixel-level probability maps to flag low-confidence regions for clinical review.
- Built a scalable validation pipeline over **1,500 CT scans (125 patients)** to measure prediction variance and quantifying model reliability in the presence of anomalies.

Warren Buffett Chatbot Agent

Jun. 2025 – Aug. 2025

AI Engineer (Python, OpenAI API, LangChain)

- Built an end-to-end conversational AI system using **LangChain** and **OpenAI models**, achieving **95%** response accuracy in simulating Warren Buffett–style investment reasoning.
- Integrated external financial and search APIs to support real-time retrieval and multi-tool reasoning, enabling robust handling of **100+** distinct query types.